

Homework 4

MATH 518 - Fall 2025

Due on Wednesday, November 12, 11:59pm on Crowdmark

Total of points = 30 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic zero.

1. (10 points) Let X, Y be quasi-projective varieties, and $f: X \rightarrow Y$ a map. Suppose that $X = \bigcup_{i \in I} U_i$ and $Y = \bigcup_{i \in I} V_i$ are two covers of X and Y by open subsets $U_i \subseteq X$ and $V_i \subseteq Y$, such that $f(U_i) \subseteq V_i$ for all $i \in I$.
 - (a) Show that f is a morphism if and only if each restriction $f_i := f|_{U_i}: U_i \rightarrow V_i$ is a morphism.
 - (b) If U_i and V_i are affine varieties, show that $f_i: U_i \rightarrow V_i$ is a morphism if and only if $f_i^*(k[V_i]) \subseteq k[U_i]$.

Solution:

(a) Suppose that f is a morphism. Then

1. f is continuous *i.e.* for any open $U \subseteq Y$, the preimage $f^{-1}(U) \subseteq X$ is open,
2. for any open subset $U \subseteq Y$, we have $f^*k[U] \subseteq k[f^{-1}(U)]$.

Let us show that the same holds for the restrictions f_i . If $U \subseteq V_i$ is an open subset of V_i , then $U = U_i \cap U'$, for some open set $U' \subseteq Y$. Then

$$f_i^{-1}(U) = f_i^{-1}(V_i \cap U') = f_i^{-1}(V_i) \cap f^{-1}(U') = U_i \cap f^{-1}(U') \quad (1)$$

is open in U_i , since $f^{-1}(U')$ is open in X by the continuity of f . (Note that here we use that f is the same as f_i on the set U_i). This shows that f_i is continuous.

For the second part, suppose that $g \in k[U]$. Since U is open in V_i , it is open in X and

$$f^*(g) \in k[f^{-1}(U)] = \bigcap_{p \in f^{-1}(U)} \mathcal{O}_p(X),$$

since f is a morphism. On the other hand, the pullback of $f_i^*(g)$ is $f^*(g)|_{U_i}$, the restriction of $f^*(g)$ to U_i . Thus we have

$$f_i^*(g) \in \bigcap_{p \in f^{-1}(U) \cap U_i} \mathcal{O}_p(X) = k[f^{-1}(U) \cap U_i].$$

Conversely, let's suppose that each f_i is a morphism. We begin by showing that U is continuous. So suppose that $U \subseteq Y$ is an open subset. Then for each $i \in I$, the set $U \cap V_i$ is open in V_i . Since f_i is continuous, we have

$$f_i^{-1}(U \cap V_i) = f^{-1}(U) \cap f_i^{-1}(V_i) = f^{-1}(U) \cap U_i$$

is open in U_i , thus also open in X , since U_i is open in X . It follows that

$$f^{-1}(U) = \bigcup_{i \in I} f^{-1}(U) \cap U_i \subseteq X$$

is open, being a union of open sets. For the second part, we have $U = \bigcup_{i \in I} U \cap V_i$ with each $U \cap V_i$ being open in V_i . Since f_i is a morphism, we have

$$f_i^*k[U \cap V_i] \subseteq k[f_i^{-1}(U \cap V_i)] = k[f^{-1}(U) \cap U_i].$$

The pullback $f^*(g)$ of a $g \in k[U]$ is a function on $f^{-1}(U)$. The restriction of $f^*(g)$ to U_i is $f^*(g)|_{U_i} = f_i^*(g) \in k[f^{-1}(U) \cap U_i]$. So $f^*(g)$ is regular on each $f^{-1}(U) \cap U_i$ for each $i \in I$, and we deduce that

$$f^*(g) \in \bigcap_{i \in I} k[f^{-1}(U) \cap U_i] = k[f^{-1}(U)].$$

This shows that $f^*k[U] \subseteq k[f^{-1}(U)]$.

- (b) Without loss of generality we can assume that $U_i \subseteq k^n$ and $V_i \subseteq k^m$. One direction is clear: if f_i is a morphism then $f_i^*(k[V_i]) \subseteq k[U_i]$, by the definition of a morphism. For the converse direction, suppose that $f_i^*(k[V_i]) \subseteq k[U_i]$. For $j \in \{1, \dots, m\}$, let $x_j: k^m \rightarrow k$ be the projection map on the j -th component, and $\bar{x}_j$ the restriction to V_i . It defines an element in $k[V_i]$ (which can be viewed as the element $\bar{x}_j \in k[V_i] \simeq k[x_1, \dots, x_m]/I(V_i)$). By our assumption, we have $f_i^*(\bar{x}_j) \in k[U_i]$ is regular. But $f_i^*(\bar{x}_j) = f_i^{(j)}$ is the j -th coordinate of the map f_i . Thus $f_i = (f_i^{(1)}, \dots, f_i^{(m)})$, where each $f_i^{(j)}$ is regular on U_i . This shows that f_i is a polynomial map, so a morphism of affine varieties.

2. (10 points) Let $f: X \rightarrow Y$ be a morphism of quasi-projective varieties. Let $X' \subseteq X$ and $Y' \subseteq Y$ be closed subvarieties such that $f(X') \subseteq Y'$. Show that the restriction

$f|_{X'}: X' \longrightarrow Y'$ is a morphism.

Hint: Show it for affine varieties first, then use the previous exercise

Solution: First, suppose that $X \subseteq k^n$ and $Y \subseteq k^m$ are affine varieties. If f is a morphism, then there exists a polynomial map

$$F = (F_1, \dots, F_m): k^n \longrightarrow k^m, \quad F_i \in k[x_1, \dots, x_n]. \quad (2)$$

such that the restriction of F to X gives f . Thus, it is clear that the restriction $f|_{X'}$ is also a morphism, since it is the restriction of F to X' .

To deduce the statement for general varieties, let $X = \bigcup_{i \in I} U_i$ and $Y = \bigcup_{i \in I} V_i$ be open affine covers (so each U_i resp. V_i is open in X resp. in Y , and isomorphic to an affine subset of k^n resp. k^m). By taking intersections if necessary, we can assume that the sets are small enough that $f(U_i) \subseteq V_i$. By the previous exercise, each restriction $f_i: U_i \longrightarrow V_i$ is a morphism, in this case a morphism of affine varieties. Now let $X' = \bigcup X' \cap U_i$ and $Y' = \bigcup Y' \cap V_i$. These are also open affine covers of X' and Y' . Moreover, since X' and Y' are closed, we have that $X' \cap U_i \subseteq U_i$ is closed in U_i , and $Y' \cap V_i \subseteq V_i$ is closed in V_i . We are now exactly in the previous affine situation, and we deduce that the restriction

$$f|_{X' \cap U_i}: X' \cap U_i \longrightarrow Y' \cap V_i \quad (3)$$

is a morphism. Using the previous exercise again, this time in the other direction, we deduce that $f: X' \longrightarrow Y'$ is a morphism.

3. (10 points) Let $X = k^2 \setminus (0, 0)$.

(a) Show that $k[X] = k[x, y]$.

(b) Deduce that X is not isomorphic to an affine variety.

Hint: Use the fact that if Y is an affine variety, then any proper ideal $I \subset k[Y]$ defines a nonempty set $V(I)$, by the Nullstellensatz.

(c) Deduce that the union of two open affine subvarieties of a variety is not necessarily affine.

Solution:

(a) Let $V = k^2$. We have $X = U_x \cup U_y$ where $U_f = \{p \in V \mid f(p) \neq 0\}$. Hence, we have

$$k[X] = k[U_x] \cap k[U_y].$$

We have seen in class that

$$k[U_x] = k[V] \left[\frac{1}{x} \right] = \left\{ \frac{g(x, y)}{x^n} \mid g(x, y) \in k[x, y] \right\},$$

$$k[U_y] = k[V] \left[\frac{1}{y} \right] = \left\{ \frac{h(x, y)}{y^m} \mid h(x, y) \in k[x, y] \right\}.$$

Thus, if f in $k[X]$ then

$$f|_{U_x} = \frac{g(x, y)}{x^n}, \quad f|_{U_y} = \frac{h(x, y)}{y^m}$$

and on the intersection $U_x \cap U_y$ we have

$$f|_{U_x \cap U_y} = \frac{g(x, y)}{x^n} = \frac{h(x, y)}{y^m},$$

which implies that

$$y^m g(x, y) = x^n h(x, y)$$

for all $(x, y) \in U_x \cap U_y$. If two polynomials agree on a dense subset, they agree everywhere. Indeed, suppose that $f, g \in k[x, y]$ are two polynomials that agree on a dense subset $U \subseteq k^2$, i.e. $f(p) - g(p) = 0$ for all $p \in U$. Then $U \subseteq V(f - g) \subseteq k^2$, and by taking the closure we deduce that $V(f - g) = k^2$, which implies that $f = g$.

Thus, $y^m g(x, y) = x^n h(x, y)$ for all $(x, y) \in k^2$, and the polynomials are equal (since we are in characteristic 0). It follows that y^m divides $h(x, y)$, that x^n divides $g(x, y)$ and that

$$f = \frac{g(x, y)}{x^n} = \frac{h(x, y)}{y^m} \in k[x, y].$$

- (b) Suppose we had an isomorphism $\varphi: W \rightarrow X$, where $W \subset k^n$ is some affine variety. The induced isomorphism of k -algebras is

$$\varphi^*: k[X] = k[x, y] \rightarrow k[W].$$

Let $\mathfrak{m} \subset k[x, y]$ be the maximal ideal $\mathfrak{m} := (x, y)$, and $I := \varphi^*(\mathfrak{m}) \subset k[W]$ its pullback. Since $\mathfrak{m}$ (and thus I) is a proper ideal, the algebraic subset

$$V(I) := \{q \in W \mid f(q) = 0 \text{ for all } f \in I\}$$

is nonempty. Since φ is a bijection, the set $\varphi(V(I))$ should be nonempty as well. But

$$\begin{aligned} \varphi(V(I)) &:= \{p \in X \mid 0 = f(\varphi^{-1}(p)) = (\varphi^{-1})^*(f)(p) \text{ for all } f \in I\} \\ &:= \{p \in X \mid 0 = g(p) \text{ for all } g \in (\varphi^{-1})^*(I) = \mathfrak{m}\} \end{aligned}$$

is empty, since the $V(\mathfrak{m}) = (0, 0)$ is not included in X . We obtain a contradiction, and deduce that such an isomorphism φ does not exist.

- (c) The previous example shows that $k^2 \setminus \{(0, 0)\} = U_x \cup U_y$. Both are affine, but the union is not.